CHAPTER 24

Wildlife Toxicity Assessment for Benzo[a]Pyrene

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Contents

INTRODUCTION

Benzo[a]pyrene (B[a]P) (CAS No. 50-32-8) is one of a large number of polycyclic aromatic hydrocarbons (PAHs) that are formed during the incomplete combustion of organic matter. Though structurally related, the compounds display considerable heterogeneity in the number and spatial arrangement of their fused aromatic rings. Mixtures of PAHs are ubiquitous in the air because of stack emissions, smoke from household wood and coal fires, cigarettes, barbecue grills, automobile exhausts, and so on. Fossil fuels are also a major source of PAHs, and this is of relevance to military sites. The compounds may be secondarily dispersed to other environmental media. Seventeen PAHs including B[a]P have been identified as being especially important for environmental monitoring and compliance purposes [1]. These can be further identified by whether or not they have been demonstrated or are thought likely to induce or promote tumor formation in experimental animals. Benzo[a]pyrene is the most well-studied PAH, a

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cancer-inducing substance for which toxicological data serve as quantitative benchmarks for the entire carcinogenic subgroup. Thus, although toxicological data on the carcinogenicity of other PAHs are poor, quantitative estimates of their carcinogenic potency can be expressed relative to that of B[a]P using toxicity equivalency factors (TEFs) [2]. The carcinogenic potency of B[a]P (defined as increased risk per mg/kg/day) was first estimated using data from several animal test systems. The geometric mean determination was estimated at 3.1 mg/kg/day. The cancer potency for a number of related PAHs was estimated and then each was rounded off to a simple multiple (1.0, 0.1, 0.01, or 0.001) of the B[a]P value. PAHs always occur in complex mixtures, and the TEF approach may be used to provide an aggregate assessment of cancer risk, rather than focusing on B[a]P alone.

Much of the experimental data on the toxicity of B[a]P is irrelevant to the development of wildlife Toxicity Reference Values (TRVs) for the compound, since their primary focus is on tumor formation. In general, TRVs are based on noncancer effects, of which there is paucity of studies for B[a]P. The TRVs presented are intended to serve as benchmarks for screening-level ecological risk assessments. This Wildlife Toxicity Assessment (WTA) summarizes available information on the likely effects of B[a]P on wildlife. The TRVs are documented in the U.S. Army Center for Health Promotion and Preventive Medicine Technical Guide 254, *Standard Practice for Wildlife Toxicity Reference Values* [3].

TOXICITY PROFILE

Environmental Fate and Transport

There is an extensive bibliography on the dispersion of PAHs such as B[a]P in environmental media, with HSDB [4] and ATSDR [1] providing summaries of the available information. The fact that B[a]P has been detected in all environmental media and in foodstuffs [4] attests to the compound's ability to become widely dispersed when released to the atmosphere, and to the comparative ease with which it can become taken up into the food chain.

The ATSDR [1] discusses the potential for B[a]P and other PAHs to become dispersed throughout environmental media following their release to the atmosphere in combustion emissions. Natural sources of PAH emissions include forest fires and volcanoes, while anthropogenic sources include the residential burning of wood; generation of industrial power; incineration of waste; production of coal tar, coke and asphalt; and exhaust emissions from use of fossil fuels, as in automobiles, among many others. Typically, a complex mixture of PAHs will be released by the processes of which B[a]P is but a single, and not necessarily the most heavily represented, component.

Airborne PAHs undergo widespread dispersion in the atmosphere but will ultimately deposit in surface water or soil because of wet or dry deposition. In surface water, the compounds can volatilize, photolyze, oxidize, biodegrade, bind to suspended particles or sediments, and/or bioaccumulate in aquatic organisms. However, the combined low water-solubility and vapor pressure, and high octanol-water partition coefficient (K_{ow}) of B[a]P suggest that the compound will partition to the soil compartment primarily, with sediment secondary, and air, water, and biota much less heavily favored [1]. Table 24.1 summarizes the physical and chemical properties of B[a]P.

The capacity for PAHs to undergo abiotic degradation in the atmosphere, for example, by photooxidation, is governed largely by the physical nature of the compound. Compounds absorbed to soot are more resistant to photochemical reactions than pure compounds; however, photolysis may be an important process by which B[a]P molecules in water can be degraded. A number of microbial genera are able to break down PAHs to related but not necessarily simplified structures. In addition, some algae and fungi are

Table 24.1 Summary of the Physical-Chemical Properties of Benzo[a]pyrene

CAS No.	50-32-8
Molecular weight	252.32
Color	Pale yellow
Physical state	Monoclinic crystals/plates
Melting point	179 °C
Boiling point	310–312 °C (at 10 mm Hg)
Odor	Faint aromatic
Solubility in water	1.6–2.3 µg/L at 25 °C: miscible with hydrocarbon solvents
Partition coefficients:	
Log K_{ow}	5.97–6.06
Log K_{oc}	6.74
Vapor pressure at 25 °C	$5.5 - 5.6 \times 10^{-9}$ mm Hg
Henry's Law constant at 25 °C	4.9×10^{-7} atm.m^3/mole
Conversion factors	1 ppm = 10.32 mg/m^3 1 mg/m^3 = 0.097 ppm

Sources: ATSDR [1], HSDB [4].

effective, with biodegradation being an important process by which PAHs can be removed from soils and sediments [1].

Summary of Mammalian Toxicity

Mammalian Oral Toxicity

By far, the majority of research on the toxicity of B[a]P has involved mammalian test species largely as a surrogate for human exposure and toxicity. The vast majority of these studies, therefore, are of little utility for wildlife TRVs. Other than a few key research articles that characterized effects of B[a]P on TRV-relevant endpoints (e.g., survival, growth, reproduction), most articles focus on various elements of the carcinogenic mechanism and manifestation of B[a]P. With the advent of the "omics" revolution for example, a wide array of studies have been produced that are more mechanistically oriented.

Mammalian Oral Toxicity: Acute

There are no listed values for the median lethal dose (LD_{50}) of B[a]P via the oral route in such secondary references as IARC [5], ATSDR [1], Lewis [6], and HSDB [4]. However, many studies have explored the acute toxicity of B[a]P to experimental animals. For example, Reddy *et al.* [7] studied the induction of nuclear anomalies in the gastrointestinal tract of B6C3F1 mice gavaged with a single dose 192 mg/kg of B[a]P (and other PAHs) in dimethyl sulfoxide. All experimental animals were sacrificed 24 to 48 hours after dosing, and the incidence of nuclear anomalies was monitored in the forestomach epithelium by quantitative histopathology/cytology. By using a range of PAHs of differing carcinogenic potential, the authors concluded that the induction of nuclear anomalies by the tested compounds was broadly aligned to their relative carcinogenic potency in the gastrointestinal tract. However, in general, cancer studies are not considered highly relevant to a WTA.

Saunders *et al.* [8] evaluated the effect of a single oral dose of B[a]P in male rats. The endpoints included neurobehavioral effects as well as antioxidant enzyme levels and lipid peroxidation in selected brain regions. Animals were given a single oral dose (25–2,000 mg/kg) of B[a]P and motor activity was determined at 2, 4, 6, 12, 24, 48, 72, and 96 hours posttreatment. Benzo[a]pyrene treatment markedly affected motor activity of males that were generally restricted to early observation periods and higher concentrations of the chemical. Treatment with B[a]P also affected brain enzyme levels that suggested the involvement of oxidative stress in neurobehavioral effects in a mechanism that was partly mediated by inhibited antioxidant scavenging systems of the brain [8]. Although the lowest observed effect level was 50 mg/kg, the effects were not observed beyond 48 hours, which suggested recovery from the initial exposure.

Knuckles *et al.* [9] evaluated the acute toxicity of B[a]P to F-344 rats. An equal number of rats/sex/group were dosed once by oral gavage with 0, 100, 600, or 1,000 mg benzo[a]pyrene/kg in peanut oil. Rats were sacrificed 14 days later, following which blood specimens were taken to quantify a number of hematological parameters, and organs (e.g., ovaries, liver, stomach, kidney, and testes) were weighed and evaluated for morphological and histopathological analysis. The liver-to-body-weight ratio was significantly lower in male and female treated animals at 100 mg/kg and 600 mg/kg, respectively, as compared to controls. The white blood cell counts and mean cell hemoglobin concentrations were also markedly lower in the males only at 600 mg/kg as compared to the controls.

The study by Reddy *et al.* [7] illustrated the propensity of PAHs like B[a]P to cause profound site-specific lesions when administered to experimental animals (e.g., at doses from 48 to 192 mg/kg/day), for which the development of skin papilloma in the nude mouse skin-painting protocol is the best documented example. However, several reports showed that at least a portion of the applied dose traverses the absorption barrier that could also be transported to remote sites. Kleisch *et al.* [10] studied chromosome damage of mouse bone marrow by B[a]P administered to (101 × C3H) F_1 mice in single oral doses ranging from 0 to 250 mg/kg. Chromosome analysis of bone marrow specimens containing 2,000 polychromatic erythrocytes at 12, 30, or 54 hours postdosing showed treatment-related increases in micronuclei and in a dose-dependent manner.

Benzo[a]pyrene can cause both target-specific lesions and effects in distant organs or tissues that result from migration of B[a]P from the site of exposure. Benzo[a]pyrene occupies the hydrocarbon hydroxylase receptor (AHH), and similar to observations with dioxins, can elicit various toxic metabolites. Simultaneously, this leads to the breakdown of PAHs, thereby limiting bioaccumulation and decreasing the overall toxic load.

Mammalian Oral Toxicity: Subacute

Anselstetter and Heimpel [11] studied the subacute effects of B[a]P and development of hematotoxicity in DBA/2 and BDF1 mice. Severe bone marrow depression, with near complete destruction of pluripotent hematopoietic stem cells, was seen in female DBA/2 mice after administration of oral B[a]P (125 mg/kg body weight (bw)/day) for 13 days. Extreme resistance to bone marrow toxicity was observed in BDF1 mice fed for 19 days [11].

Mammalian Oral Toxicity: Subchronic

Ramesh *et al.* [12] carried out a subchronic study in F-344 rats exposed to B[a]P in the diet at concentrations equivalent to doses of 0, 5, 50, or 100 mg/kg for 90 days. Five animals/sex/group were sacrificed after 30, 60, and 90 days, and liver microsomal preparations were monitored for aryl hydrocarbon hydroxylase (AHH) activity. Both dose- and time-dependent increases in AHH activity were obtained, potentially contributing to the formation of toxic reactive metabolites and disease symptoms in target organs such as the liver.

Knuckles *et al.* [9] evaluated the subchronic toxicity of B[a]P to F-344 rats for 90 days. Forty rats/sex/group were given 0, 5, 50, and 100 mg/kg/day of B[a]P administered via feed. The actual doses received (consumed) by the animals were approximately 10% of the nominal doses; that is, 0.5, 5, and 10 mg/kg bw/day, respectively. The effects of B[a]P on body weight, hematological parameters, and organ mass, gross alteration, and histopathology were also studied. The diet was a mixture of lab meal that was obtained from the Purina Ralston Company and the respective B[a]P concentration. The liver, kidney, stomach, prostate, testes, and ovaries were weighed and evaluated for gross and histopathological changes. Data obtained from hematological analyses and evaluation of the organs was analyzed by two-way ANOVA, Fisher's Exact Test, and the Cochran-Armitage test.

At 90 days, Knuckles *et al.* [9] found that the red blood cells (RBC) and hematocrit levels had significantly decreased at 50 mg/kg/day for males and 100 mg/kg/day for females. Hemoglobin levels had significantly decreased for males and females at 100 mg/kg/day.

Wolford *et al.* [13] established reference blood parameter data for mice and rats. The study presented typical blood parameter values observed in mice and rats. Although Knuckles *et al.* [9] states that RBC, hematocrit, and hemoglobin levels had significantly decreased at particular doses at 90 days, these values fall within the normal range [13], indicating that the findings were not biologically significant.

No significant differences between the control and experimental groups regarding body weight were reported. Tubular casts in the kidney tissues were reported to be abnormal from the controls in males at 50 and

100 mg/kg/day after 90 days. Eighty percent of the males were affected at the 50 mg/kg/day level and 100% of the males were affected at 100 mg/kg/day. Ten percent of the females at 50 mg/kg and 100 mg/kg were found to have abnormalities in the tubular casts in the kidneys [9].

DeJong *et al.* [14] studied the immunotoxicological effects of B[a]P to male Wistar SPF rats. Eight rats/group were dosed via gavage with 3, 10, 30, or 90 mg benzo[a]pyrene/kg bw in soybean oil. After 35 days of exposure, the rats were necropsied and body weight, blood, and organs were evaluated. The immunotoxicological parameters studied were bone marrow, thymus, spleen, and lymph nodes. The adrenals, brain, bone marrow, colon, caecum, jejunum, heart, kidney, liver, lung, lymph nodes, esophagus, pituitary, spleen, stomach, testis, and thymus were evaluated for gross and histopathological changes. All organs were weighed with the exception of the bone marrow, pituitary, and GI tract.

The immunotoxicological parameters and organ weights were assessed using one-way ANOVA. The critical values in the blood data were determined by using the protocol per Statistical Tables and Biometry and The Principles and Practice of Statistics in Biological Research [15,16]. The histopathological data were evaluated using Fisher's exact test.

The results in the study stated that there was a significant difference in body weight between the control and the 3 mg/kg dose groups. This was true in the 90 mg group but not in the intermediate 10 and 30 mg groups. There was a significant difference in RBC, hemoglobin, and hematocrit between the control and the 10 mg/kg dose groups. Although all values in the DeJong *et al.* [14] study were statistically significant, the findings are not biologically significant. The RBC levels fall within a normal range as defined previously [13]. The hemoglobin levels in the DeJong *et al.* study were slightly higher [14] than the values stated in Wolford *et al.* [13] yet are not considered biologically significant since the increase was not that much higher compared to the reference value. The hematocrit value was higher than the reference value in Wolford *et al.* [13], although it was not considered biologically significant since the other blood parameters, including the RBC and hemoglobin levels, fell within the normal range, and it was likely that since the RBC and hemoglobin levels increased, the hematocrit levels were also increased.

Mammalian Oral Toxicity: Chronic

No studies on the chronic toxicity of B[a]P were located.

Mammalian Oral Toxicity: Other

MacKenzie and Angevine [17] conducted an experiment to examine the reproductive and developmental effects of oral B[a]P exposure. The study assessed the effects of B[a]P to CD-1 mice upon exposure during the fetal stage. Thirty to 60 pregnant female mice/group were gavaged with 0, 10, 40, or 160 mg/kg bw in 0.2 mL corn oil on gestation days (GDs) 7 to 16, then allowed to deliver and nurse their young. After treatment and giving birth to pups, the litter size, pup weight, and percentage of viable litters were recorded. Litters (F_1) were allowed to remain with their mothers until postnatal day 20, at which point the dams were removed. At 7 to 8 weeks old, each F_1 male was housed with two untreated virgin females every 5 days for 25 days. Pregnant females from these matings were sacrificed on GD 19, and the number of implants, fetuses, and resorptions was recorded. The F_2 young were examined for gross abnormalities. At 7 to 8 weeks old, each F_1 female was housed with untreated proven breeder males for periods of up to 6 months. All resulting F_2 young were counted and examined for gross abnormalities on day 1 of life, then sexed and weighed at 4 days of age.

The data on litter size for F_1 and F_2 populations were analyzed using the least significant difference method. The data on pup weight and the fertility index in the F_1 males and females were analyzed using nonparametric statistics. There was no significant difference between the control and all experimental dose groups regarding the mean litter size and percentage of viable litters in the F_1 population. There was a statistically significant decrease in mean pup weight in all B[a]P exposed groups (F_1 population). A significant decrease in fertility index (females pregnant/females exposed to males $\times$ 100) had resulted in the F_1 males and females at 10, 40, and 160 mg B[a]P /kg/day. F_2 mean litter size by F_1 females was significantly lower at 10 mg/kg/day compared to controls. The results indicate that the effects of B[a]P exposure during development had manifested in adult reproduction for the F_1 population. These results indicate that a prenatal exposure of 10 mg/kg/day is an appropriate lowest observable adverse effect level (LOAEL).

Kristensen *et al.* [18] reported that B[a]P might suppress development of primordial oocytes during fetal life. Benzo[a]pyrene was given to female NMRI mice, 10 mg/kg/body weight by oral intubation on days 7 to16 of gestation (F0 generation). The female pups (F1 generation), thus exposed prenatally to B[a]P, showed markedly reduced fertility with few ovarian follicles compared to controls.

Gao *et al.* [19] evaluated the effects of orally and interperitoneally administered B[a]P on female mouse survival, growth, and cervix morphology.

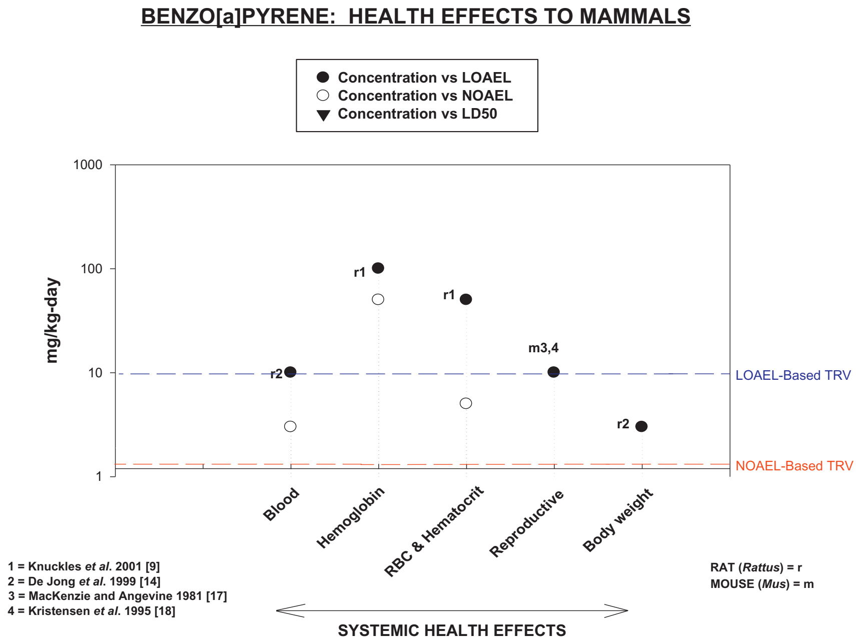

Figure 24.1 Benzo[a]Pyrene: Health Effects to Mammals. Letters by symbols represent test species, and numbers represent study.

Female mice received either an oral or interperitoneal dose of 0, 2.5, 5, or 10 mg B[a]P/kg body weight, twice a week for 14 weeks. B[a]P treatment caused a variety of effects including increased enzyme activity of CYP1A1, creatine kinase, and aspartate aminotransferase (AST), and decreased activity of glutathione-S-transferase in the cervix. Cervix morphology was altered and weight was reduced in B[a]P treated mice and 5.0 and 10 mg B[a]P/kg significantly decreased survival and body mass in female mice. Overall, IP injection of B[a]P caused more severe effects than did oral gavage. There were no significant effects at 2.5 mg B[a]P/kg body weight. Table 24.2 summarizes the relevant mammalian oral toxicity data for TRV derivation

Studies Relevant for Mammalian TRV Development for Ingestion Exposures

Although the majority of studies on B[a]P focus on carcinogenic effects, several studies are potentially suitable for deriving a mammalian TRV. Ramesh *et al.* [12] found that there was a dose-dependent and time-dependent increase in AHH activity. However, due to the lack of histopathological analyses to determine if increased levels of AHH had an adverse effect on the liver, this

Table 24.2 Summary of Relevant Mammalian Oral Toxicity Data for TRV Derivation

Test Organism	Test Duration	Test Results: NOAEL (mg/kg/day)	LOAEL (mg/kg/day)	Effects Observed at the LOAEL	Study
Mice (CD-1)	GDs 7–16	ND	10	Reproductive effects (fertility, decreased birth weight)	[17]
Mice (NMRI)	GDs 7–16	ND	10	Decreased ovarian follicles, litters, litter size	[18]
Rats (Wistar SPF; males)	35 days	ND	3	Decreased body weight, decreased growth	[14]
		3	10	Increased RBC, hematocrit, and hemoglobin	
Rats (F-344)	90 days	5	50	Increased RBC and hematocrit levels (males)	[9]
		50	100	Increased hemoglobin levels	

GD, gestation day; ND, not determined; RBC, red blood cell(s).

study will not be used to derive a mammalian TRV. Knuckles *et al.* [9] and DeJong *et al.* [14] assessed the toxic effects of B[a]P during subchronic exposures. Knuckles *et al.* [9] found that there was no significant effect of B[a]P on body weight whereas DeJong *et al.* [14] found a significant decrease in body weight in B[a]P exposed rats. The blood parameter data in both studies were not biologically significant as determined by comparisons with reference data in Wolford *et al.* [13]. The only endpoint suitable for TRV development in the aforementioned studies was the altered growth of B[a]P exposed rats (at the lowest level, 3 mg, and at the highest level, 90 mg/kg, but not at the 10 mg or 30 mg levels), as seen by DeJong *et al.* [14]. This study, however, was not chosen for TRV development because of the biological relevance of the statistical 3 mg/kg LOAEL finding and because the MacKenzie and Angevine [17] study was considered more suitable because it was a multigenerational reproductive study with clear effect-based endpoints identified.

MacKenzie and Angevine [17] conducted a developmental/reproductive study showing significant reproductive effects of B[a]P in mice. The study assessed the effects of B[a]P on the development of F_1 pups exposed during gestation, and when they matured, on the reproduction of the F_1 population. Importantly, although this study is not chronic in the classic sense (i.e., study duration), the exposure occurred during a sensitive life stage (development), so it was considered equivalent in weight of a chronic exposure regime [3]. Pregnant dams exposed to B[a]P at 10 mg/kg/day from GD 7 to 16 had given birth to F_1 pups with significantly lower birth weights. Subsequently, when the F_1 mice (exposed to 10 mg/kg/day during development only) reached adulthood, females produced significantly smaller, or no (at higher doses) litters and males and females had a significantly lower fertility index indicating that the effects of prenatal exposure to B[a]P were evident even in adulthood at the lowest experimental dose (10 mg/kg/day). As a result, the data regarding fertility index will be used to derive the mammalian TRVs. This result, with regard to female mice of the NMRI strain, is supported by the findings of Kristensen *et al.* [18], in which a LOAEL of 10 mg/kg/day was established for reduction in litters and litter size and in ovarian follicles. As outlined in TG254, endpoints used to derive a TRV should have a clear link to an ecologically relevant effect [3]. For B[a]P, the reduction in reproductive output and the apparent long-lasting effects of prenatal exposure shown by MacKenzie and Angevine [17] has clear implications for the sustainability of exposed mammalian populations.

Mammalian Inhalation Toxicity

As summarized in ATSDR [1], several inhalation studies exist that address the carcinogenicity of B[a]P via the inhalation route, especially those resulting in site-of-impact lesions to the respiratory tract. Greenwood *et al.* [20] conducted an acute inhalation study pertaining to noncarcinogenic effects of B[a]P emphasizing the neurotoxicological effects at the molecular level in Sprague Dawley rats. The maternal rats were exposed to a dose of B[a]P: carbon black aerosol (100 $\mu g/m^3$) via inhalation on GD 15 for 4 hours. An Sp1 transcription factor consensus sequence was examined by electrophoretic mobility shift analysis of nuclear extracts from various brain regions in pups on postnatal days 3, 5, 7, 10, and 15. The results showed changes in the developmental expression of Sp1 abundance. The data obtained on the temporal and spatial regulation of gene expression in the brain indicate that an effect of the transplacental deposition of desorbed B[a]P to the fetus is in-utero neurotoxicity.

Archibong *et al.* [21] exposed pregnant F-344 rats to B[a]P via the inhalation route at 25, 75, and 100 $\mu g/m^3$ for 4 hours a day for 10 days (GDs 11–20) and evaluated effects on their offspring. Fetal survival decreased in a dose-dependent manner from 78.3 to 33.8% from lowest to highest exposure levels compared to nearly 100% survival in controls. There were also significant decreases in plasma progesterone, estrogen, and prolactin in exposed females associated with B[a]P exposure.

Archibong *et al.* [21] also evaluated effects of inhaled B[a]P on intratesticular function in F-344 rats. Rats were exposed to 75 $\mu g/m^3$ for 4 hours a day for 60 days. At day 60 and then the following 24, 48, and 72 hours, plasma testosterone and luteinizing hormone were measured as indicators of testicular function. Testis weight, total tubule mass, and total tubular length decreased 33, 27, and 39% compared to controls and the number of homogenization-resistant spermatids was markedly reduced in B[a]P exposed rats; plasma testosterone was significantly lower and LH significantly higher in exposed rats as well. Taken together, the data suggest 60-day exposures to B[a]P contributed to reduced endocrine and spermatogenic function.

Wormley *et al.* [22] evaluated the neurotoxic effects in the F1 generation when parental generation were exposed to B[a]P via inhalation. Dams were exposed on GDs 11–21 for 4 hours per day and maintained to term and pups weaned on post-partum day (PD) 30. Electrophysiological studies were conducted PD 60–70 and revealed impacts to long-term potentiation, a cellular correlate of memory and learning.

Ramesh *et al.* [23] evaluated the effect of inhaled B[a]P on fertility endpoints in male F-344 rats. Rats were exposed to 75 μg B[a]P/m^3 for 4 hours a day for 60 days while controls were unexposed. Blood samples were collected after study cessation and again at 24, 48, and 72 hours. Inhaled B[a]P caused reductions in testis weight, components of spermatogenic and steroidogenic compartments of the testis, and progressive motility and mean density of stored spermatozoa were also significantly reduced. There was also a decrease in plasma testosterone and an increase in luteinizing hormone in B[a]P exposed rats. The authors concluded that subchronic exposure to B[a]P via inhalation reduced testicular and epididymal function in rats.

Mammalian Dermal Toxicity

No studies were found.

Mammalian Toxicity from Uncommon Exposure Pathways

There were a variety of studies on B[a]P toxicity in which the exposure route evaluated could be considered uncommonly studied. With growing apparent concerns about the effects of PAHs on humans, a number of studies that evaluated effects of parental exposure on fetuses and newborns were identified. Although in most cases, the endpoints evaluated were generally subtle and not likely relevant for TRV development. Nonetheless, several are highlighted to provide perspective on current environmental toxicity of B[a]P. Bouayed *et al.* [24] evaluated the effects of lactational exposure to B[a]P on postnatal neurodevelopment, gene expression, and behavior. Overall, they found changes in neurodevelopment, gene expression, and behavior (assessed using different mazes) in mice whose mothers were given oral doses of either 2 or 20 mg B[a]P/kg for 14 days and there were significant effects of B[a]P on neonatal mice at both exposure levels.

Summary of Avian Toxicology

Hoffman and Gay [25] found that B[a]P causes embryotoxicity in avian species. Their study design consisted of surface applications of a mixture containing B [a]P to Mallard Duck eggs during a critical period of development (72 hours of development). Proportions of B[a]P (i.e., 0.02, 0.10, or 0.50%) were combined with a petroleum hydrocarbon mixture that was applied to the egg surface [25]. The B[a]P mixture of 0.02% resulted in a significant decrease in embryonic growth and increased occurrence of abnormal survivors. Levels of 0.1% B[a]P resulted in a significant decrease in survival and more abnormal survivors than the 0.05% B[a]P group. Levels of 0.5% B[a]P resulted in mortality of all animals [25]. Abnormal survivors were characterized as having bill defects, gastroschisis, incomplete ossification, stunted embryos with incomplete feather formation, eye and brain defects, anophthalmia, and exencephaly.

Others have assessed the embryotoxicity of B[a]P in Chicken Hens *Gallus domesticus*, and determined mean weight change, food intake, egg production, eggshell thickness, liver weight, and hepatic microsomal proteins after a 5-day treatment with 5 mg of B[a]P that was dissolved in 0.25 mL corn oil and placed in gelatin capsules. The hens were orally administered B[a]P capsules without any marked differences in the measured endpoints [26]. A related study tested B[a]P for embryotoxicity in Chicken Hen *Gallus domesticus* embryos [27]. The study design consisted of injecting 2.0 mg/kg B[a]P into egg yolks that were preincubated for 4 days, following which embryonic mortality was measured 2 weeks later [27]. The vehicle was peanut oil, lecithin, and water. Thirty percent embryonic mortality had resulted, a significant difference from the control. None of the data in the avian studies will be used to derive avian TRVs. Chen *et al.* [25] did not find any significant differences, and the egg injection study by Brunstrom *et al.* [27] and an egg surface application study by Hoffman and Gay [25] do not accurately reflect real-life exposures for birds.

Amphibian Toxicology

Studies designed to obtain more information on the carcinogenic effects of B[a]P have been conducted [28–32]. To date, no amphibian studies have evaluated the noncarcinogenic health effects of B[a]P.

Reptilian Toxicology

There are little to no data on the toxicity of B[a]P to reptiles. In fact, the only studies identified were focused more on polycyclic aromatic hydrocarbons as a whole. One study [33] was designed to improve understanding of the potential effects of polycyclic aromatic hydrocarbons (PAHs) on the development of snapping turtles (*Chelydra serpentina*) residing in a PAH contaminated wildlife refuge. Turtle eggs from clean reference sites were exposed to a crude oil mixture but also included a B[a]P only exposure. There were limited effects of B[a]P on development except for turtles from one reference site where there was a linear relationship between deformity severity and B[a]P exposure concentration. Exposure levels are difficult to determine because exposure mixtures were administered to the eggshell. Nonetheless, these are among the only data that indicate a potential adverse effect of B[a]P on turtle hatchlings. Marsili *et al.* [34] also demonstrated that Italian wall lizards (*Podarcis sicula*) could prove useful as bioindicators of adverse ecological effects for habitat near oil fields. They linked several biomarkers (enzymic activity of P450 1A1; PAH metabolites) with exposures to a suite of chemical stressors, which included PAHs. While neither study was suitable for TRV development, the studies suggest potential for adverse ecological effects to reptiles [33] and that exposures in the field may occur [34].

RECOMMENDED TOXICITY REFERENCE VALUES

Toxicity Reference Values for Mammals

TRVs for Ingestion Exposures for the Class Mammalia

Results from a reproductive study on mice [17] showed significant effects of B[a]P on the fertility index in male and female F_1 mice and on mean litter size in female F_1. These reproductive effects have clear implications for the sustainability of exposed populations; thus, this study is ideal for the derivation of a mammalian TRV. The LOAEL for effects on the fertility index was 10 mg/kg/day, which was the lowest dose tested. The MacKenzie and Angevine [17] study was suitably designed and executed, and was equivalent to chronic exposure data, since exposures occurred during a sensitive life

Table 24.3 Selected Ingestion TRVs for the Class Mammalia

TRV	Dose (mg/kg/day)	Confidence Level
NOAEL-based	1.0	Medium
LOAEL-based	10	Medium

Source: [17].

stage [3]. Since the dose-response data does not include a no observable adverse effect level (NOAEL), the TRV was derived using the NOAEL-LOAEL approach, and the results are presented in Table 24.3. An uncertainty factor of 10 was used to derive the NOAEL-based TRV from the LOAEL [3]. The TRVs are the same for both males and females since the exposure effects on fertility were similar. A medium confidence rating was applied to these TRVs since the study was appropriately designed and representative of a chronic study. Too few mammalian orders were represented, which precluded a high confidence TRV.

TRVs for Inhalation Exposures for the Class Mammalia

Not available at this time.

TRVs for Birds

Not available at this time.

TRVs for Amphibians

Not available at this time.

IMPORTANT RESEARCH NEEDS

The limited availability of data on the toxicity of B[a]P to wildlife species precludes the development of a high-confidence TRV. Hence, more studies on the toxicity of B[a]P to wildlife species are needed. Particularly warranted are long-term, chronic toxicity studies on mammals and additional studies on nonmammalian wildlife such as birds, reptiles, and amphibians. Although there are a number of studies on the effects of B[a]P on mammalian species, many of these are of poor quality and cannot be used to generate TRVs. Further experimentation, with close attention to experimental design, would greatly increase confidence in B[a]P TRVs.

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